II. Answer any 10 questions (including Question No.17 compulsory) in few sentences.

10 X 3 =30

16. Define Ohm's law.
17. If 100 Ohms and 150 Ohms are connected in parallel in a circuit, calculate the total resistance of the circuit.
18. What is a transformer?
19. What is atomic weight?
20. What is called half-wave rectifier?
21. Write short notes on SMD transistor.
22. How will you display number 3 in seven segment display?
23. What is meant by piezo electric effect?
24. What are the three basic logic gates?
25. Why digital system is reliable?
26. Define the working of ROM & RAM.
27. Define computer.
28. Write the uses of Logic Probe.

III. Explain any 5 questions (including Question No.31 compulsory) in a Paragraph

5 X 5 =25

29. Draw and explain about electrolytic capacitors.
30. Explain Dynamic microphone with neat diagram.
31. Draw the circuit diagram of bridge rectifier and explain its working function
32. Explain the working principles of NPN transistor
33. Explain Binary number system.
34. Write short notes on (i) Keyboard (ii) Plotter (iii) Hard disk (iv) Pen Drive
35. List out any five functions of a multimeter

IV. Explain in about a page.

2 X 10 = 20

36. (a) In a series circuit, two resistors 150 Ω and 100 Ω are connected with 125 V supply. Prove that the sum of the voltage drops across the resistors is equal to the EMF applied in the circuit.

(OR)

(b) Explain the working principle of RC phase shift oscillator with circuit diagram.

37. (a) Explain the construction and working of an IGBT

(OR)

(b) Explain the working of CPU with neat diagram.

ELEC_EQUIP_Chapter10.indd 228 11/10/2020 11:55:39 AM